

CHARACTERISTICS OF SCRIPTING LANGUAGE

- **Integrated Compile and Run**

Scripting Language behave as if they were interpreted. They are often an interactive, experimental activity that doesn't fit well with the "edit-compile-link-run" cycle of conventional programming.

A few Scripting Languages like Unix Shell and TCL(Tool Command Language) version 7 are indeed implemented as strict interpreters.

However most of the languages in current use opt for hybrid technique which involve compiling to an intermediate form which is then interpreted.

- **Low Overhead and ease of use**

Based on the usage variables can be declared and initialized . Later focus was on explicit variable declaration, which requires that you declare every variable before you use it.

Eg: Dim sum As Integer (or) sum=10;

- **Enhanced Functionality**

Most of the languages provide powerful string manipulation based on the use of regular expression(A regular expression is a string of characters that define the pattern or patterns you are viewing.). Some languages support easy access to API or Object model exported by application.

CHARACTERISTICS OF SCRIPTING LANGUAGE

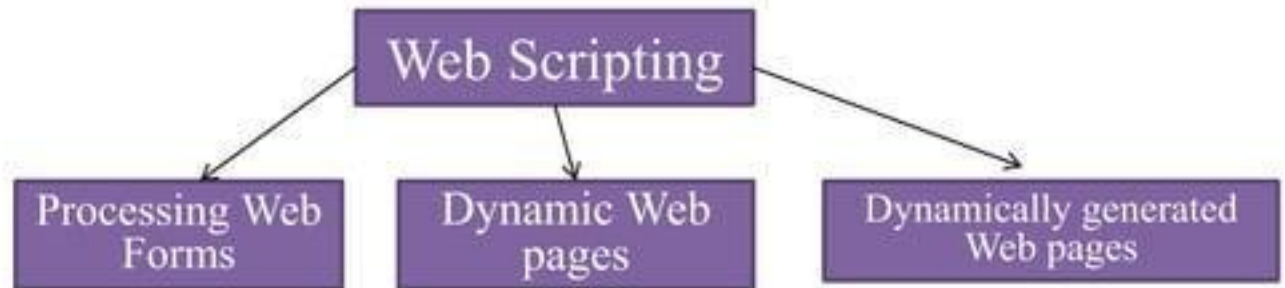
- **Efficiency is not an issue**

Ease of use is achieved at the expense of efficiency

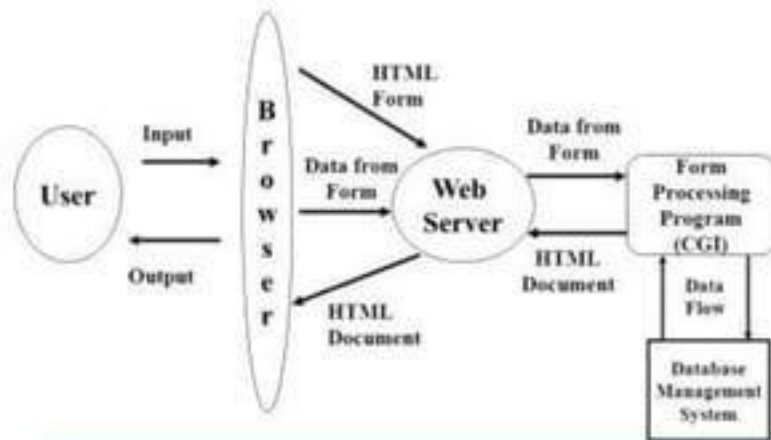
eg: interpretation rather than compiling

Focus is not on high performance but on the speed of development together with ability to make changes to meet new requirement.

- Interpreted languages also have a simple syntax which, for the user: makes them easy to learn and use.
- assumes minimum programming knowledge or experience.
- allows complex tasks to be performed in relatively few steps.
- allows simple creation and editing in a variety of text editors.
- allows the addition of dynamic and interactive activities to web pages.
- Also, interpreted languages are generally portable across various hardware and network platforms and scripts can be embedded in standard text documents for added functionality.



- Processing Web Forms



User Registration

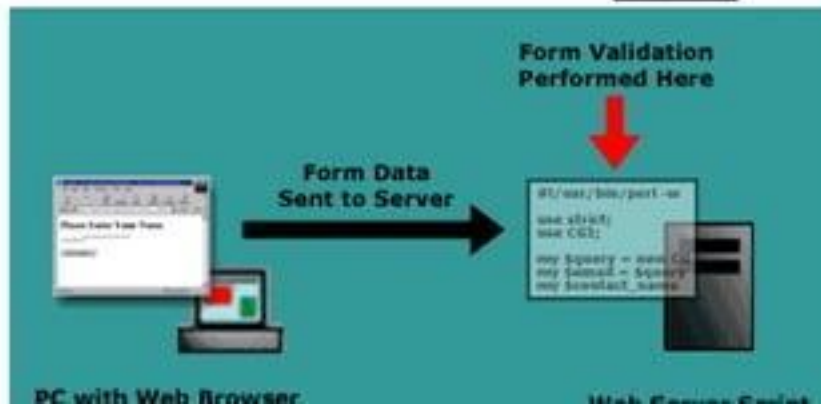
First Name
Please enter your firstname

Last Name
Please enter your lastname

Email
Please enter a valid email address

Password
Please provide a password

By clicking register you agree to our [policy](#).



DYNAMIC WEB PAGES

- DHTML is about creating web pages that reacts to (user)events.
- DHTML is about using JavaScript and the HTML DOM to change the style and positioning of HTML elements.

```
<html>
<head>
<script type="text/javascript">
function message()
{
alert("This alert box was triggered when you selected the content of the input field");
}
</script>
</head>

<body>

<p>Select the content in the input field</p>
<form>
<input type="text" value="Select this text" onselect="message()" size="20">
</form>

</body>
</html>
```

This alert box was triggered when you selected the content of the input field

☐ Prevent this page from creating additional dialogues.

OK

Select the content in the input field

Select this text

Dynamically generated HTML

- It is another form of dynamic web page is one in which some or all of the HTML is generated by scripts executed on the server.
- A common application is to construct a page whose content is retrieved from a database.
- Examples: 1) Microsoft IIS Web Server which implements Active Server Pages(which has scripts in java script or vb script)

```
<%@ page import = "java.sql.*" %>
<%
    try{
        String username = request.getParameter("username");
        String password = request.getParameter("password");
        Class.forName("com.mysql.jdbc.Driver"); // MySQL database connection
        Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/jav?" + "user=root&password=");
        PreparedStatement pst = conn.prepareStatement("Select user,pass from login where user=? and pass=?");
        pst.setString(1, username);
        pst.setString(2, password);
        ResultSet rs = pst.executeQuery();
        if(rs.next())
            out.println("Valid login credentials");
        else
            out.println("Invalid login credentials");
    }
    catch(Exception e){
        out.println("Something went wrong !! Please try again");
    }
}%>
```

Login Here	
Username	
Password	
Please fill in this field.	
Login	Reset

UNIVERSE OF SCRIPTING LANGUAGES

- There is scripting universe, containing multiple overlapping worlds
- The original UNIX world of traditional scripting using Perl and Tcl
- The Microsoft world of visualbasic and active x controls.
- The world of VBA for scripting compound documents.
- The world of client-side and server-side web scripting.
- Overlapping of them is very complex, as Web scripting can be done in VBScript,JavaScript,Perl or Tcl.
- **Perl and Tcl are used to implement complex application for large organization.**
 1. **Eg. Tcl has been used to develop major banking system.**
 2. **Perl is used to implement enterprise wide document management system for leading aerospace company.**



A screenshot of a software window titled "Banking Application". The window has a menu bar with "File" and "Account". Below the menu bar, it displays "Current Balance: 5411". There are two input fields: "Deposit:" with the value "1234" and "Withdraw:". At the bottom, a status bar shows "Deposited 1234."

Banking Application	
File Account	
Current Balance: 5411	
Deposit:	1234
Withdraw:	
Deposited 1234.	



TYPICAL USES OF SCRIPTING LANGUAGES INCLUDING JAVASCRIPT INCLUDE:

Image or text rollovers: If the user rolls the mouse over a graphic or hypertext then a text or graphic box will appear:

Creating a **pop-up window** to display information in a separate window from the Web page that triggered it. This is useful if the user requires to perform a simple calculation or consult a calendar for inputting dates. This is achieved by embedding ActiveX controls or Java applets into the script.

Validating the content of fields: When filling in forms, each field, especially required fields denoted by an asterisk, are validated for correct input. If the field is left blank or incorrect information entered then a user message will be generated and you may not continue.

Please provide the following details. This is necessary in case we need to contact you regarding delivery of your order.

*Title:

* First Name:

*Last Name:

*Email Address:

*Phone Number:

If you would like your order delivered to someone other than yourself, please enter their contact details below.

*Title:

* First Name:

*Last Name:



Pop-up Calander

USES OF SCRIPTING LANGUAGES

- Scripting languages are of two kinds:
 - 1.Traditional Scripting
 - 2.Modern Scripting

➤ **Traditional Scripting:**

The activities which require traditional scripting include

1. System administration
2. Controlling Remote Applications
3. System and Application extensions
4. Experimental programming
5. Command Line interface

➤ **Modern Scripting**

1. Visual Scripting
2. Scriptable components
3. Client-side and Server-side Web scripting